

Adam Matthew Steinberger

Software Engineer · AI systems, open source and technical writing

I build AI software whose claims can be checked: the design is a paper, the behavior is pinned by tests, and the code is open.

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SUMMARY

For 14 years I've built production software, and lately I've written it up as carefully as I build it. vibey's design is a paper, its decisions are 86 architecture decision records, its documentation is a book, and its chaos test passes only if no job is lost or run twice. I also wrote a plain-English book on RAG chatbots for decision-makers, the Security-First Scrum framework and its training manuals, and I've mentored junior developers on three projects. The junior developer I trained alongside a 20-service platform now owns it.

HIGHLIGHTS

- **Design written as a paper.** vibey's design is published as a paper (not peer reviewed), backed by 86 architecture decision records and documentation written as a book.
- **Claims pinned by tests.** A chaos test runs 500 jobs on 8 workers, crashes a fifth of them mid-job, and passes only if no job is lost or run twice; CI holds 100% branch-coverage floors.
- **Teaching.** Trained the junior developer who now owns a 20-service platform, wrote the Security-First Scrum framework and training manuals, and mentored junior developers on three projects.
- **Open and licensed.** vibey is MIT-licensed, résumé content is CC BY 4.0, and anyone can volunteer.

EXPERIENCE

The Vibey Project (open source) · Greenville, SC

Aug 2026 – Present

Creator and maintainer

- **vibey** (MIT; on PyPI as vibey-engine). A conductor that carries a change from a spec interview through design, build and review, handing work between Claude Code, OpenAI Codex, Cursor Agent, Google Antigravity and local open-weight models, and asking a person only for the decisions that are theirs to make. AI coding agents build it, and they answer to its own CI gates.
- **Nothing is lost when an agent dies.** Every decision, finding and handoff is a row in an append-only PostgreSQL ledger. Workers claim jobs under SKIP LOCKED leases, and a handoff to another vendor must pass a model-free no-loss check or it retries, escalates or waits for a person. A chaos test runs 500 jobs on 8 workers, crashes a fifth of them mid-job, and passes only if **no job is lost or run twice**.
- **Held to its own gates.** 100% branch-coverage floors in CI, 86 architecture decision records, and a Helm chart (KEDA, kopf operator) installed on minikube in CI. In its first two months: 596 merged pull requests and 25 releases. The design is a paper; the documentation is a book.

The Vizius Group · Greenville, SC

Sep 2025 – Aug 2026

Senior Azure and AI Development Engineer

- **AI payroll automation platform** (co-lead, ~420k lines). Twenty microservices across four human-approved phases, with the final submission modeled as irreversible; RAG over the document store, AI-directed spreadsheet corrections and earnings review. I owned the Terraform, 20 Helm charts, GitOps and 10 CI/CD workflows, backed by 585 test modules. The architecture was **production-ready at day 45**, and the junior developer I trained in parallel now owns it.
- **Technical report platform** (lead, ~54k lines). Turns electrical-testing instrument data into standards-aware client reports: mail-webhook ingestion with per-document fan-out, a multi-vendor parser seam, a deterministic deficiency analyzer with LLM review, and blocking data-quality gates. Added SAML 2.0 and Entra dual-issuer SSO, replaced a shared API key with per-user bearer auth, and wrote the SOC 2 readiness assessment, STRIDE threat model and ADRs. **Ended silent false-success deploys.**
- **AI governance gateway** (sole architect, ~54k lines). One OpenAI-compatible API in front of Azure AI, Anthropic, OpenAI, Cursor, Grok and Gemini, with per-project allowlists and fallback policy, Redis rate limits, per-call cost attribution with hard spend caps, and an HMAC-signed, hash-chained, write-once audit trail. Callers authenticate with Entra ID over workload identity, so no API keys sit in the path; agent sandboxing, egress policy and SSRF checks map to the OWASP LLM Top 10 and NIST AI RMF. **Three product teams moved onto it** and retired their app-held credentials.
- **Shared platform library.** The firm's Python library for logging, alerting, dead-letter consumers and the outbox pattern, through three major versions; **adopted by 17+ repositories** and four platforms. It lives on in the open as vibey-bootstrap, part of vibey.
- **Also.** A multi-system ticket relay (N-way sync, 653 tests, 93% coverage, mypy --strict); a multi-tenant observability portal; five architecture document sets, about 180 pages; the *Security-First Scrum* framework and training manuals; mentoring junior developers on three projects.

Adam Matthew Steinberger LLC · Greenville, SC

Mar 2025 – Aug 2025

Senior Software Engineering Consultant

- **Codebase review.** 59,000 lines across 190+ files reviewed in 10 hours; found 5% test coverage and missing auth middleware, and delivered a technical brief, executive summary and phased roadmap.
- **Two RAG chatbots, 30 days each.** One fully self-hosted for a non-profit (Mistral-7B, FAISS and vLLM, with Grafana and Prometheus on every token and no external dependencies); one cloud-based for a sales agency (Gemini RAG with API-driven web search).

The Apologist Project · Remote

Apr 2026 – Present

Volunteer Software Architect

- **Project Excite.** An adapter-based relay service that hands people from an AI chat to live volunteers on Chatwoot or EchoGlobal: an explicit session state machine with idempotent teardown, Redis-backed sessions, HMAC-verified webhooks and QStash-queued delivery. Shipped as split PR stacks with security hardening (DOMPurify, CORS allowlist, Sentry PII off, rate limiting).

Lima One Capital · Greenville, SC

May 2023 – Feb 2025

Senior Software Engineer

- **Rearchitected the core integration layer** from legacy MuleSoft APIs into NestJS microservices (gRPC and REST) on PostgreSQL, with ETL pipelines and connectors across HubSpot, SharePoint, Snowflake and Salesforce.
- **Built Snow Portal**, a Snowflake job scheduler that replaced Alteryx at a fraction of the cost; full-stack .NET and React work on a mortgage-broker platform (credit-report integrations, pricing-engine APIs).

EARLIER EXPERIENCE

Transcat, Senior Software Engineer (2022–2023). Led the team delivering .NET Web APIs and a React front end for lab-equipment calibration. · **LeaseTrack**, Senior Software Engineer (2021–2022). Python and AWS Textract document parsing; a Java Spring Boot annotation system feeding an ML training pipeline. · **Akmazio Software**, Founding Engineer (2020–2021). Built the full C#/.NET and SQL Server backend and wrote the business plan; managed interns and a contractor. · **Bestpass by Fleetworthy**, Software Engineer (2019–2020). Introduced automated unit testing to a legacy toll-billing system that had none. · **New York State Insurance Fund**, Software Engineer (2015–2019). Migrated VB6 systems to C# MVC, refactored Oracle EDI integrations and mentored junior engineers. · **Town and Country Computer Services**, Junior Software Engineer (2013–2015). Insurance quoting, rating and reporting apps used all day by underwriters; client-facing from the first day. · **GE HealthCare**, Junior Software Engineer (2012–2013). A browser-based CT/MRI viewer with real-time 3D scrolling; built its full internationalization feature.

SKILLS

AI & LLM systems: Multi-vendor LLM gateways with cost and policy governance, agent sandboxing and egress policy, multi-agent orchestration, MCP servers, RAG (pgvector, FAISS, Azure AI Search, Pinecone), human-in-the-loop gating, structured outputs; Claude, Azure OpenAI/Foundry, GPT, Gemini, Mistral, vLLM, Ollama

Quality: pytest, Hypothesis, mutation and chaos testing, contract and end-to-end tests, mypy --strict, ruff, import-linter-enforced onion architecture

Languages: Python 3.12 (FastAPI, SQLAlchemy 2, Pydantic, kopf), TypeScript/NestJS, Next.js + React, C#/.NET, Java Spring Boot, SQL, KQL, Bash, gRPC/REST, OpenAPI

Data & integration: PostgreSQL, Snowflake, SQL Server, Oracle, MongoDB, Redis; ETL and API integrations (HubSpot, SharePoint, Salesforce, Microsoft Graph); event-driven pipelines (Service Bus, Event Hubs); data-quality gates

Platform & cloud: Private AKS (workload identity, KEDA), Terraform, Bicep, Helm, Kustomize, Flux/Argo CD GitOps, GitHub Actions, Service Bus, Event Hubs, Key Vault, Private Endpoints, App Gateway + WAF, OpenTelemetry, PostgreSQL, Redis, Cosmos DB, AWS

OPEN SOURCE & PUBLICATIONS

- **vibey.** Conductor for AI coding agents, five engine runners, the vibey-gh release tool, vibey-bootstrap and krypton apps for desktop, mobile, web and VS Code. MIT; pip install vibey-engine. Contributors welcome.
- **vibey-skills.** A 149-plugin Claude Code skills marketplace, shipped inside vibey.
- **Ledger-Mediated Orchestration: Vendor-Independent Autonomous Software Delivery over a Pool of Coding Agents.** Paper describing vibey's design (not peer reviewed).
- **Novice to Navigator: Your Guide to AI Chatbots for Business.** Plain-English guide to RAG chatbots for decision-makers; first edition free online (ISBN 979-8274310628).

EDUCATION & CERTIFICATION

Skidmore College, B.A., Computer Science (2010–2012) · **Rensselaer Polytechnic Institute**, Electrical and Electronics Engineering coursework (2008–2010) · **Certified ScrumMaster (CSM)**, Scrum Alliance (2021)